Advancing GPS Spoofing Detection for Autonomous Vehicles through Deep Learning and Hybrid Architectures

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Abstract: This research investigates the evolving threat of GPS spoofing within Global Navigation Satellite Systems (GNSS) and its implications for autonomous vehicle (AV) navigation. Through a comprehensive comparative analysis of three leading deep learning models Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Transformer architectures, the study evaluates each model's capacity for detecting spoofed signals under dynamic, real-world conditions. Civilian GPS signals, lacking encryption and authentication, remain highly vulnerable to adversarial manipulation, posing systemic risks to transportation, defense, and industrial infrastructures. By synthesizing literature across GNSS security, machine learning, and cyber-physical system resilience, this research demonstrates that hybridized deep learning models, particularly Transformer–CNN–LSTM frameworks, achieve superior detection accuracy (≥95–99%) with minimal false alarm rates. The findings underscore the necessity of integrating multi-sensor fusion, explainable AI (XAI), and edge computing for real-time spoofing mitigation. Ultimately, the study provides a benchmark for developing adaptive, scalable, and trustworthy GPS spoofing detection systems that align with NIST cybersecurity frameworks and ensure operational integrity for next-generation autonomous vehicles.

Keywords— GPS Spoofing Detection; Autonomous Vehicles; Deep Learning; GNSS Security

Introduction

On December 5, 2011, a landmark incident underscored the critical importance of Global Navigation Satellite System (GNSS) signal security and integrity. During a surveillance operation over Iran, a U.S. stealth drone was allegedly captured by Iranian forces through an act of electronic hijacking—specifically, Global Positioning System (GPS) spoofing (bbc.com, 2011). According to an Iranian engineer interviewed by The Christian Science Monitor, the attack succeeded by transmitting counterfeit GPS signals that misrepresented the drone's altitude and coordinates, deceiving the onboard navigation system into descending and crash-landing in Iranian territory (Peterson & Faramarzi, 2011). While U.S. officials attributed the event to a mechanical malfunction, the incident drew global attention to the vulnerabilities of GPS-dependent systems. The former U.S. Secretary of Defense, Leon Panetta, later emphasized that the ability to disrupt drone operations represented a growing cybersecurity threat with serious implications for military and civilian technologies.

This event exemplifies the geopolitical and technical risks associated with unprotected GNSS signals. Civilian GPS transmissions, which lack encryption and authentication, are particularly vulnerable to spoofing attacks that deliberately broadcast false signals to deceive receivers (Ghanbarzade & Soleimani, 2025). Spoofing occurs when radio frequency (RF) waveforms mimic legitimate satellite signals closely enough to mislead receivers, thereby denying, degrading, or distorting operational performance (U.S. DHS, 2017). The effects of such attacks can be immediate or delayed and may persist even after spoofing activity ceases. In autonomous vehicles (AVs), spoofing-induced disorientation can lead to catastrophic navigation failures, posing grave risks to passengers, infrastructure, and surrounding traffic environments.

The increasing reliance on GPS for autonomous navigation highlights an urgent need to secure these systems from hostile interference, particularly spoofing and jamming (Ghanbarzade & Soleimani, 2025). Civilian GPS signals are unencrypted and unauthenticated, allowing attackers to manipulate positioning data and misdirect AV systems (Abrar et al., 2024). One common metric for identifying spoofed signals is the carrier-to-noise density ratio (C/N_0), which measures signal strength. Spoofed signals tend to be stronger and more uniform because they originate from nearby terrestrial transmitters rather than distant satellites, resulting in a distinct and detectable power profile. Interference as little as 24 dB above the authentic signal can disrupt GPS reception (U.S. DHS, 2017). Although mitigation methods such as blocking and redundant antennas can reduce susceptibility, they do not eliminate the threat. Consequently, recent advancements in machine learning (ML) and deep learning offer promising avenues for developing adaptive and accurate spoofing detection frameworks (Ghanbarzade & Soleimani, 2025).

As self-driving services such as Waymo and Cruise expand operations across metropolitan areas, the probability of cyberattacks targeting AV fleets will continue to rise. These vehicles' dependence on weakly secured GPS signals makes them prime targets for ransomware and other cyber-physical attacks, particularly during major public events like the 2028 Olympic Games in Los Angeles, California (Ghanbarzade & Soleimani, 2025). The convergence of urban mobility technologies, high network connectivity, and critical public exposure necessitates immediate advancements in GPS spoofing detection and mitigation to safeguard transportation infrastructure. Strengthening AV navigation integrity is therefore not only a technical requirement but a public safety imperative.

Importance of Autonomous Vehicles in U.S. Critical Infrastructure

Autonomous vehicles play a vital role across numerous sectors of U.S. critical infrastructure, extending far beyond consumer ride-hailing services. Applications include autonomous buses for public transit, heavy machinery for construction and mining, unmanned aerial and ground vehicles for defense operations, automated forklifts for logistics, and autonomous aircraft and ships for aerospace and maritime navigation (U.S. DHS, 2022). Each of these applications relies heavily on GNSS positioning data, meaning that GPS spoofing poses risks not only to individual assets but to the continuity and safety of national infrastructure systems.

Table 1. AV Applications Mapped to NIST Critical Infrastructure Sectors

Industry	AV Applications/Usage	NIST Critical Infrastructure Sector Relationship
Transportation & Logistics	Self-driving trucks	Transportation Systems – Secure mobility & logistics
Ride-sharing & Mobility	Autonomous taxis & shuttles	Transportation Systems – Urban transit integration
Public Transit	Autonomous buses, harvesters	Food & Agriculture – Ensuring secure food production
Construction & Mining	Autonomous heavy machinery	Critical Manufacturing – Protecting Industrial Processes
Healthcare	Autonomous ambulances and medical delivery services	Healthcare & Public Health – Securing emergency response
Military & Defense	UAVs, autonomous combat vehicles	Defense Industrial Base – Securing strategic ops
Retail & Delivery	Autonomous delivery vehicles	Transportation Systems/Commercial – Secure supply chain
Manufacturing & Warehousing	Automated forklifts, transport vehicles	Critical manufacturing – Secure production and inventory
Aerospace & Maritime	Autonomous aircraft & ships	Transportation Systems – Secure navigation & transport

Table 1 demonstrates the intersection of autonomous vehicle applications with National Institute of Standards and Technology (NIST) Critical Infrastructure sectors. It highlights the cross-sectoral dependence on secure and reliable navigation systems, underscoring the cybersecurity implications of GPS spoofing for national defense, manufacturing, healthcare, and transportation networks. As AV technologies proliferate across these domains, ensuring robust signal authentication and spoofing detection becomes central to national resilience and operational security.

GNSS/GPS Security Fundamentals

The Global Navigation Satellite System (GNSS), with the Global Positioning System (GPS) as its most prevalent and widely adopted constellation, serves as the foundation for modern positioning, navigation, and timing (PNT) infrastructure. These systems are indispensable to autonomous vehicle (AV) operations, providing precise geolocation and temporal synchronization for navigation, obstacle avoidance, and real-time decision-making (Dasgupta et al., 2024). However, despite their critical role in supporting both civilian and defense applications, most GNSS implementations—particularly those reliant on civilian GPS signals—lack inherent encryption, authentication, and signal verification mechanisms. This structural vulnerability exposes GPS-dependent systems to a range of adversarial manipulations, including spoofing, meaconing, and signal replay attacks (Johansson et al., 2025).

GPS spoofing represents a sophisticated threat in which an adversary transmits counterfeit satellite signals to deceive a receiver into calculating false position, velocity, or timing (PVT) data (Chen et al., 2024). Because autonomous systems rely heavily on GPS inputs for operational safety, these falsified signals can produce catastrophic outcomes, such as route deviations, navigation loss, or full control compromise (Kamal et al., 2021). In a typical spoofing scenario, an attacker initially broadcasts signals that are indistinguishable from legitimate satellite transmissions. Over time, the attacker gradually increases the power of the counterfeit signals and introduces subtle positional offsets. This progression leads the receiver to lock onto the falsified signals, resulting in an undetectable drift from the true position (Pham & Xiong, 2021). The attack's stealth derives from the seamless synchronization between the spoofer's transmissions and authentic satellite signals, a factor that significantly complicates real-time detection and mitigation (Dasgupta et al., 2024).

The deceptive nature of GPS spoofing is further compounded by the synchronization phase, during which counterfeit signals align with legitimate ones before overpowering them. This transition occurs without noticeable disruptions, allowing the attacker to manipulate the receiver's computed coordinates, speed, and timing information without triggering error thresholds or alarms (Kamal et al., 2021). Such attacks often exploit the deterministic structure of civilian GPS signals, which lack cryptographic protection, making them inherently susceptible to replication and controlled modification. Once synchronization is achieved, a spoofer can execute gradual positional shifts that are nearly impossible to distinguish from ordinary signal variations, thereby misleading autonomous navigation systems into dangerous maneuvers or false geolocation fixes.

The implications of these vulnerabilities extend beyond individual vehicle security. Compromised GPS signals threaten the broader ecosystem of intelligent transportation systems, emergency response coordination, and critical infrastructure operations that depend on accurate PNT data. For example, in an urban mobility context, a single spoofing attack could redirect an entire fleet of autonomous vehicles, disrupt traffic flow, or compromise safety-of-life services. The systemic reliance on weakly authenticated signals underscores the pressing need for resilient detection algorithms, adaptive learning frameworks, and signal authentication protocols capable of distinguishing between genuine and counterfeit transmissions.

Recent research has emphasized the growing potential of machine learning (ML) and deep learning architecture such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Transformer-based models—to address these challenges. These models are designed to detect irregular signal behaviors, such as abnormal Doppler shifts, time delays, and power anomalies, by analyzing complex temporal and spatial patterns in GNSS data (Dasgupta et al., 2024). By learning from historical and simulated attack data, ML-driven detection systems can dynamically adapt to evolving spoofing tactics, offering a scalable and data-centric defense strategy against adversarial interference. The integration of such intelligent models into GNSS receivers and autonomous vehicle navigation systems represents a critical advancement in safeguarding operational integrity and enhancing trust in GPS-dependent infrastructures.

GNSS Architecture and Signal Structure

A comprehensive understanding of the Global Navigation Satellite System (GNSS) architecture and signal composition is essential for analyzing the vulnerabilities that adversaries exploit during spoofing attacks. The GNSS framework comprises a constellation of satellites that continuously transmit radio signals containing timing and positioning information. These signals enable receivers to compute their precise geolocation through trilateration—calculating distance based on the travel time of signals from at least four satellites. Within this architecture, the Global Positioning System (GPS) is the most widely utilized constellation, serving as the global standard for navigation and timing across transportation, military, and industrial domains (Modas et al., 2020).

Each GPS satellite transmits signals composed of three principal components: a carrier wave, pseudorandom noise (PRN) code, and navigation data. The carrier wave defines the radio frequency on which the signal is transmitted, while the PRN code provides a unique sequence that allows receivers to identify individual satellites and measure signal travel time. The navigation message encodes essential orbital parameters, satellite clock corrections, and system timing information that collectively enable position computation (Burbank et al., 2024). By cross-referencing multiple signals, a GPS receiver determines its three-dimensional position and precise timing offset relative to the satellite network.

The civilian GPS L1 C/A (Coarse/Acquisition) signal, which operates at 1575.42 MHz, is unencrypted and openly accessible. While this accessibility fosters widespread civilian use, it simultaneously introduces a fundamental security weakness—its unprotected structure allows adversaries to replicate or alter signals with relative ease (Gupta et al., 2023). Spoofers can exploit this vulnerability by generating counterfeit signals that imitate legitimate PRN sequences and navigation data, deceiving receivers into accepting false positional inputs. This replication is facilitated by the predictable nature of the C/A code, making it possible for attackers to synchronize their fake transmissions with genuine satellite signals. As a result, spoofed signals can seamlessly override authentic ones, leading to erroneous navigation outputs without triggering conventional error-detection mechanisms (Mykytyn et al., 2023).

Modern spoofing attacks leverage advanced hardware capable of high-fidelity signal generation and precise timing control, allowing for near-perfect emulation of authentic satellite transmissions. A well-calibrated spoofer can simultaneously broadcast multiple counterfeit satellite signals, each with correct PRN codes, realistic Doppler shifts, and consistent timing characteristics, to fully simulate an authentic GNSS environment (Zhou & Han, 2023). By maintaining synchronization with legitimate signals before gradually diverging, the attacker achieves a "carry-off" effect—misleading the receiver into transitioning from genuine to counterfeit signals in a gradual and nearly imperceptible manner (Pham & Xiong, 2021).

Even less sophisticated spoofing methods, such as meaconing, can effectively manipulate GPS receivers. Meaconing involves capturing authentic GNSS signals, delaying them slightly, and rebroadcasting them at higher power. Despite its simplicity, this technique can cause significant navigation errors, as the receiver interprets the delayed signals as originating from a greater distance, leading to inaccurate position and time calculations (Iudice et al., 2024). Although accurate positioning typically requires simultaneous reception from at least four satellites, high-precision applications often depend on six or more (Raponi et al., 2021). This dependency increases the attack surface because spoofing a subset of signals can still introduce substantial positional drift or timing error (Meheretu et al., 2024).

The unique PRN codes assigned to each satellite enable receivers to differentiate signals; however, these same codes can be replicated by attackers to construct high-fidelity counterfeit transmissions. Once spoofed signals are introduced, even low-cost GNSS receivers—commonly found in consumer-grade AV systems—may fail to discern the difference, particularly in the presence of high-power local transmissions (Olesen, 2017). Moreover, advanced spoofing systems can synchronize and replay multiple PRN-coded signals in real time, effectively masking inconsistencies and eluding conventional integrity checks (Kumar et al., 2022). This manipulation exploits the inherent trust model of GNSS architecture, which assumes all signals with valid structure and timing originate from legitimate satellites.

Consequently, the GNSS signal structure—while designed for precision and interoperability—presents an exploitable weakness when faced with intentional interference. As adversarial techniques evolve, spoofers can replicate the signal environment with increasing accuracy, nullifying basic detection approaches. These vulnerabilities underscore the necessity for intelligent detection frameworks that combine signal processing and data-driven modeling. Machine learning (ML) and deep learning techniques, such as convolutional and recurrent neural networks, are now being developed to monitor temporal variations, phase coherence, and cross-satellite relationships, enabling systems to distinguish authentic signals from those exhibiting the subtle anomalies characteristic of spoofing attacks. The continued refinement of these methods represents a critical frontier in the advancement of GPS security for autonomous navigation systems.

GPS Vulnerabilities and Security Risks

Despite their critical role in global navigation and autonomous systems, GPS signals remain inherently fragile due to their low transmission power and unencrypted civilian structure. Authentic satellite signals arrive at Earth's surface with a power level typically around −130 dBm, making them highly susceptible to interference from locally generated counterfeit signals that operate at higher amplitudes (Sallouha et al., 2024). This susceptibility enables both jamming, the deliberate broadcasting of high-power noise that disrupts legitimate signal acquisition, and spoofing, in which false but coherent signals deceive receivers into computing erroneous position, velocity, or timing (PVT) information (Miralles et al., 2020). Because spoofing attacks mimic authentic signals rather than simply overpowering them, they are far more insidious, often progressing undetected until system integrity has already been compromised.

The primary vulnerability of civilian GPS systems arises from their lack of encryption and signal authentication, features that would otherwise prevent unauthorized transmission or modification. Attackers exploit this openness by synchronizing counterfeit signals with authentic ones before gradually manipulating code phase, carrier frequency, or navigation data to introduce false measurements, a process known as a carry-off attack (Wesson et al., 2017). This technique allows the spoofer to progressively mislead the receiver without triggering traditional alarms based on signal power or correlation metrics. Moreover, because low-cost GNSS receivers used in autonomous vehicles often possess limited processing power and minimal anti-spoofing capability, they are particularly vulnerable to such sophisticated adversarial techniques (Hamza et al., 2024).

Replay-based attacks present an additional layer of risk. In these scenarios, adversaries record genuine GNSS signals and later retransmit them, either synchronously or with intentional delays, to mislead receivers into calculating incorrect positions and times (Zeng et al., 2017). Even small deviations in signal characteristics, such as mismatched code or carrier phases introduced through meaconing, can generate significant positioning errors while maintaining receiver lock, thereby evading detection (Islam et al., 2024). This subtle manipulation of timing and phase values can deceive a receiver's tracking loops, causing it to follow the falsified signal path while believing it is processing authentic satellite data.

Another significant vulnerability stems from the prevalence of multipath effects, where signals reflect off surfaces before reaching the receiver. Spoofers can exploit this natural phenomenon by designing signals that imitate multipath distortions, thereby masking malicious activity. This strategy complicates spoofing detection since standard receivers are programmed to tolerate such reflections within acceptable thresholds (Wesson et al., 2017). Similarly, short-duration jamming can serve as a precursor to more complex spoofing operations, briefly overwhelming the receiver's tracking loops before counterfeit signals are introduced in a synchronized "handover" phase (Spanghero et al., 2025). Such hybrid attacks, combining jamming, meaconing, and spoofing, pose severe risks to safety-critical applications like autonomous driving, where even minor positional deviations can result in hazardous outcomes.

Compounding these risks is the increasing affordability and accessibility of spoofing technology. Commercially available software-defined radios (SDRs) and open-source GNSS signal libraries allow threat actors to build spoofing transmitters with minimal cost and expertise (Islam et al., 2024). This democratization of signal manipulation capabilities has led to a surge in GPS interference incidents globally, affecting aviation, maritime navigation, and ground-based autonomous systems (Khan et al., 2021). Furthermore, even simple spoofing devices can exploit receiver design assumptions, such as the expectation that stronger signals are inherently more reliable. Because authentic satellite signals are so weak, a modest power increase in counterfeit transmissions is often sufficient to override legitimate reception (Wesson et al., 2017).

Traditional defense mechanisms, such as power monitoring or signal-to-noise analysis, are insufficient in isolation. These methods cannot reliably differentiate between jamming, spoofing, or benign environmental interference, particularly in complex urban or industrial environments where signal reflections and power fluctuations are common (Wesson et al., 2017). As a result, modern countermeasures increasingly rely on multi-layered detection frameworks that integrate signal integrity monitoring, multi-antenna direction finding, and machine learning-based anomaly detection. Such approaches allow systems to detect subtle, multi-dimensional irregularities in signal properties, such as phase coherence, Doppler shifts, or temporal correlations that may indicate spoofing activity.

Ultimately, the convergence of low-power signal transmission, open access to waveform specifications, and widespread dependence on GPS for safety-critical applications creates a severe systemic vulnerability. The disruption of GPS signals can have cascading effects across interdependent infrastructure sectors, including aviation, energy distribution, emergency services, and financial systems. For autonomous vehicles, a single spoofing incident could lead to catastrophic navigation errors, physical collisions, or fleet-wide service disruptions. Addressing these vulnerabilities demands a combination of cryptographic authentication, intelligent detection algorithms, and adaptive machine learning systems capable of continuously retraining to detect evolving spoofing patterns in real-world environments. The implementation of these countermeasures is vital to ensuring resilient, trustworthy, and secure GPS-based navigation in the era of autonomous systems.

Spoofing Attacks: An Overview

GPS spoofing attacks represent a wide spectrum of adversarial techniques designed to deceive receivers by imitating authentic satellite signals. These attacks range from simple signal replays to highly sophisticated methods that generate fully synthetic signals with precise timing, power, and data characteristics. Unlike jamming, which merely disrupts communication, spoofing manipulates the receiver's perception of space and time, creating a false navigation environment that appears legitimate. Such attacks are particularly concerning for autonomous vehicles and other safety-critical systems that depend on continuous and accurate positioning data (Islam et al., 2024).

Sophisticated spoofers employ highly coordinated strategies that synchronize counterfeit signals with legitimate ones to avoid immediate detection. By aligning the counterfeit signals' pseudoranges, Doppler shifts, and timing sequences with those of the genuine satellites, attackers can seamlessly transition control from authentic to falsified signals (Ardizzon et al., 2023). This synchronization, achieved through meticulous calibration, ensures that the receiver perceives the transition as natural. Once synchronization is established, the attacker gradually introduces deviations in code phase and navigation message data to mislead the receiver. These subtle manipulations make detection challenging, especially for low-cost receivers with limited computational resources or basic integrity checks (Soufi et al., 2024).

Spoofing attacks can take various forms, including meaconing, carry-off, nulling, and replay attacks, each exploiting a distinct aspect of GNSS vulnerabilities. In a carry-off attack, the spoofer begins by transmitting synchronized counterfeit signals and gradually increases their power while deviating their navigation parameters, effectively pulling the receiver away from authentic signals (Wesson et al., 2017). In contrast, nulling attacks involve suppressing genuine signals while simultaneously broadcasting falsified ones from the same location. This forces the receiver to lose lock on the authentic satellites and re-acquire the stronger counterfeit signals (Wesson et al., 2017). These attacks can be executed with high precision using software-defined radios capable of generating coherent, multi-channel signals with realistic Doppler and timing profiles.

Replay or meaconing attacks are simpler but still effective. In these scenarios, attackers record authentic satellite signals and re-transmit them after a short delay or with deliberate timing shifts. This manipulation alters the receiver's perception of satellite distances, causing position errors without any visible sign of interference (Miralles et al., 2020). Even low-power meaconing can be successful if the delayed signal appears stronger than the authentic one, especially during cold starts when the receiver is reacquiring satellite data (Islam et al., 2024). Attackers can also employ ghost map techniques, which not only falsify positioning data but also alter the displayed map or navigation output, thereby masking the discrepancy between real and spoofed coordinates (Miralles et al., 2020).

The complexity of spoofing is amplified by its potential for synchronization-based deception, where false signals are carefully tuned to align with authentic transmissions in both frequency and phase. This alignment ensures that receivers cannot easily distinguish spoofed signals based on conventional power or timing metrics. Advanced spoofers can even emulate satellite motion, signal dynamics, and orbital geometry, thereby constructing a fully realistic yet fraudulent navigation scenario (Islam et al., 2024). In such cases, traditional monitoring of signal strength or correlation functions is insufficient because both authentic and counterfeit signals appear statistically consistent.

Detection of these advanced attacks requires analyzing subtle signal characteristics that go beyond power and timing metrics. Some approaches utilize angle-of-arrival analysis, temporal cross-correlation, and signal distortion monitoring to identify anomalies in propagation paths or phase coherence (Lemieszewski et al., 2021). Other methods rely on integrity monitoring frameworks that use redundant sensor data or multiple receivers to cross-validate positional information. For instance, spoofing can often be detected by comparing time and range measurements between multiple GNSS receivers located in proximity, since genuine signals exhibit predictable geometric relationships while spoofed signals do not (Spanghero & Papadimitratos, 2024).

Attackers continually refine their methods to exploit receiver limitations. Even small power advantages, sometimes as little as 0.4 dB, can cause a receiver to prioritize spoofed signals over authentic ones (Wesson et al., 2017). Additionally, nulling-and-replacement attacks, which cancel authentic signals and substitute identical replicas, remain among the most difficult to detect. These attacks require detailed modeling of the attacker-to-receiver channel and careful control of phase and amplitude parameters, making them especially effective against commercial receivers (Wesson et al., 2017). The mathematical modeling of such spoofing environments has become a key area of research, enabling the development of algorithms that identify inconsistencies in correlation envelopes, spectral content, and phase dynamics (Iudice et al., 2024).

The diversity of spoofing techniques underscores the necessity for adaptive detection frameworks that integrate multiple methodologies. Signal-processing approaches, while effective against basic spoofing, often struggle with advanced methods that mimic authentic signals with near-perfect fidelity (Truong et al., 2023). To enhance resilience, researchers are increasingly incorporating data-driven models that use machine learning to identify non-linear and high-dimensional relationships in GNSS signal data. These models can detect subtle anomalies across frequency bands, time intervals, and spatial configurations that traditional filters fail to capture. By combining statistical detection with artificial intelligence, modern GNSS receivers can significantly improve sensitivity to spoofing while reducing false alarm rates.

In summary, GPS spoofing attacks exploit inherent weaknesses in the GNSS architecture, including open signal structures, predictable codes, and limited receiver authentication. As spoofers adopt more sophisticated synchronization and nulling techniques, the challenge for researchers and engineers is to develop adaptive, multi-modal detection systems that can recognize and counter these evolving threats. The fusion of traditional signal analysis with deep learning architecture offers a promising path forward for ensuring the reliability, integrity, and security of autonomous navigation systems operating in contested electromagnetic environments.

GPS Spoofing Threats

The complexity of GPS spoofing threats extends beyond simple replication of satellite signals and encompasses a range of sophisticated methods that manipulate signal characteristics to create subtle but significant navigation errors. Modern spoofing strategies target the integrity of the Position, Navigation, and Timing (PNT) framework by exploiting vulnerabilities in signal authentication, synchronization, and receiver algorithms. Unlike jamming, which merely blocks access to navigation signals, spoofing intentionally alters the data received, thereby misleading the system into calculating false locations or trajectories without triggering conventional alarms (Wesson et al., 2017).

These attacks can deceive receivers by gradually introducing modified signals that are consistent with legitimate satellite transmissions, allowing adversaries to assume control of the navigation process. One of the most effective techniques involves power ramping or code-phase alignment, where an attacker first synchronizes counterfeit signals with authentic ones and then slowly increases their signal strength. As a result, the receiver begins tracking the counterfeit signals while maintaining apparent consistency with legitimate data (Wesson et al., 2017). Once this alignment is achieved, the attacker can introduce progressively larger deviations in position or time without immediate detection. These gradual shifts allow the attack to persist for extended periods, especially in autonomous systems that rely on continuous GPS updates for operational decisions.

Such spoofing attacks are particularly dangerous because they exploit inherent trust assumptions built into GPS receivers. Most receivers are designed to interpret the strongest, most coherent signal as authentic. Therefore, when an adversary transmits counterfeit signals with slightly higher power or better synchronization, the receiver often locks onto the spoofed transmission by default. In extreme cases, when the retransmitted signal possesses significantly higher power, it can even saturate the receiver's tracking loops, leading to permanent signal lock on falsified data (Islam et al., 2024). This manipulation can cause the receiver to report false position and time information, effectively allowing attackers to redirect, disorient, or disable autonomous vehicles, unmanned aerial systems, or maritime navigation platforms.

Advanced spoofing strategies can also simulate multipath propagation effects by introducing small, controlled variations in signal timing and amplitude. By doing so, spoofers mimic environmental reflections that typically occur in urban or obstructed environments, which are considered normal by GPS receivers. These artificial multipath effects mislead the receiver's signal-processing algorithms, which are designed to tolerate minor variations in path length. Consequently, spoofing signals that emulate natural distortions can persist undetected for extended durations (Kareem, 2024).

Spoofers may also employ coordinated multi-channel attacks targeting multiple satellites simultaneously, thereby creating an entirely fabricated navigation solution. These synchronized attacks modify several signals in parallel, making the false navigation data appear geometrically consistent across multiple satellite constellations. This type of coordinated deception is particularly effective against multi-constellation receivers that rely on signals from GPS, Galileo, or BeiDou systems for redundancy. Because all the falsified signals appear consistent, the receiver's integrity monitoring algorithms fails to recognize the attack. Such coordinated spoofing can result in the generation of false three-dimensional trajectories, leading to catastrophic outcomes in safety-critical applications such as aviation and autonomous driving.

The operational consequences of these threats are profound. In autonomous vehicles, spoofing can result in unsafe maneuvers such as unplanned lane departures, route deviations, or unintended stops. In aviation, it can lead to course misalignments and altitude errors that threaten flight safety. For maritime vessels, spoofing can redirect ships into restricted or hazardous waters. In each of these cases, the spoofing attack undermines not only navigation accuracy but also situational awareness, thereby increasing the potential for accidents, property loss, or intentional sabotage.

The availability of open-source tools and inexpensive software-defined radios has lowered the barrier to entry for executing such attacks. Threat actors no longer require military-grade equipment to carry out GPS spoofing. With basic programming knowledge and access to public GNSS signal specifications, attackers can design spoofing waveforms capable of deceiving commercial receivers. This proliferation of accessible spoofing technology has expanded the threat landscape, exposing civilian and industrial systems to risks once limited to defense and intelligence contexts.

Mitigating these threats requires a combination of multi-layered detection, signal authentication, and machine learning-based anomaly recognition. Detection systems must analyze not only received power but also Doppler shifts, phase continuity, and inter-satellite relationships to identify inconsistencies that cannot be fabricated by a single spoofer. Machine learning models trained on large GNSS datasets can identify subtle statistical irregularities that distinguish legitimate signals from synthetic ones, providing an adaptive framework for defense. For instance, deep learning architectures such as Long Short-Term Memory (LSTM) networks and Transformer-based models can detect temporal anomalies in GPS data that are often invisible to traditional rule-based methods.

The growing sophistication of GPS spoofing threats underscores the necessity for resilient, adaptive, and data-driven countermeasures. Defensive strategies must evolve continuously to counter increasingly complex spoofing methodologies that exploit both hardware and algorithmic weaknesses in GNSS receivers. As autonomous systems become more integrated into transportation, logistics, and national defense, ensuring the authenticity of GPS signals is not merely a technical requirement but a critical element of safety, security, and public trust.

Types of Spoofing Attacks

Spoofing attacks can be categorized into several distinct types, each exploiting specific weaknesses in Global Navigation Satellite System (GNSS) signal structure and receiver behavior. These categories include carry-off attacks, nulling attacks, replay attacks, meaconing, and coordinated multi-signal attacks. Each of these methods employs different strategies to mislead GNSS receivers by transmitting counterfeit signals that are indistinguishable from legitimate satellite transmissions. Understanding these variations is critical for developing effective detection and mitigation strategies for autonomous vehicles and other navigation-dependent systems (Islam et al., 2024).

The carry-off attack is one of the most common and effective forms of spoofing. In this approach, the attacker begins by transmitting counterfeit signals that are synchronized with authentic satellite signals. Once synchronization is achieved, the attacker gradually increases the power of the counterfeit signals and introduces small variations in navigation data or code phase. As a result, the receiver transitions from the genuine signals to the stronger falsified ones without immediate detection (Wesson et al., 2017). The carry-off method is particularly effective because it exploits the receiver's trust in the continuity and stability of satellite transmissions. Since the transition occurs gradually, the receiver's position solution drifts over time rather than changing abruptly, which makes the attack difficult to identify through simple power or timing analysis.

A second major category is the nulling attack, in which the spoofer deliberately cancels or suppresses authentic GNSS signals while simultaneously transmitting its own counterfeit versions (Wesson et al., 2017). This technique involves precise control of phase and amplitude to ensure that the authentic signals are neutralized at the receiver's antenna input. Once the genuine signals are suppressed, the spoofer replaces them with its own signals that contain false navigation information. Nulling attacks are especially difficult to detect because the receiver continues to function normally, perceiving the spoofed signals as authentic. The attack effectively forces the receiver to track false signals while believing it is still receiving legitimate satellite transmissions.

Another type of spoofing involves replay attacks, also known as meaconing. In these cases, an adversary records authentic GNSS signals and retransmits them after a controlled delay. This replayed signal causes the receiver to miscalculate satellite distances and timing, leading to false positional data (Miralles et al., 2020). Because the replayed signals originate from real satellites, they contain valid data and structure, making them extremely challenging to distinguish from legitimate transmissions. Meaconing attacks require relatively simple equipment and are often executed using commercially available software-defined radios. They are particularly effective in urban environments or areas with limited line-of-sight to satellites, where multipath reflections are already common (Islam et al., 2024).

A more sophisticated approach is the coordinated multi-channel attack, where the spoofer transmits multiple counterfeit signals that emulate several satellite sources simultaneously. This approach ensures geometric consistency across the fabricated signals, allowing the receiver to compute a false position, velocity, and timing solution that appears mathematically valid. The coordinated multi-channel attack is highly effective against multi-constellation and multi-frequency receivers because it can falsify entire constellations of data. This method requires advanced signal synthesis and precise timing control but has been demonstrated using modern software-defined radio platforms (Miralles et al., 2020).

In addition to these primary categories, researchers have identified several emerging variations. One such example is the ghost map attack, in which the spoofer manipulates not only the GNSS signal but also the receiver's associated navigation interface or mapping system. By synchronizing the falsified signals with a modified map display, attackers can conceal positional errors from human operators or automated systems that depend on geospatial data for decision-making (Miralles et al., 2020). Another variation is the untargeted spoofing attack, which does not rely on

precise synchronization with authentic signals. Instead, the attacker broadcasts a general set of false satellite signals into a region, causing receivers that are initializing or recovering from a cold start to acquire and lock onto the spoofed signals by default (Islam et al., 2024).
These different types of spoofing attacks vary in sophistication, intent, and required resources, but all share a common objective: to compromise the integrity and reliability of navigation systems. The diversity of techniques underscores the need for a multi-faceted defense strategy that combines physical, algorithmic, and data-driven methods. Traditional countermeasures, such as monitoring signal power, Doppler shifts, or correlation symmetry, often fail against sophisticated spoofing attacks that closely replicate authentic signal characteristics (Wesson et al., 2017). As a result, researchers are increasingly turning to machine learning (ML) and deep learning techniques that can model complex, non-linear relationships among signal parameters.

For example, deep neural networks trained on high-dimensional GNSS datasets can identify patterns of signal distortion or timing anomalies that are imperceptible to conventional statistical methods. ML-based models such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Transformer architectures have shown promise in differentiating authentic signals from spoofed ones based on temporal and spatial patterns (Truong et al., 2023). These models can analyze variations in power, phase, or spectral content to detect subtle inconsistencies across multiple satellite channels, enabling real-time spoofing detection even under complex environmental conditions.

The diversity of spoofing attack types reflects the evolving nature of GPS vulnerabilities and the sophistication of modern threat actors. Each type of attack presents unique detection challenges, requiring integrated and adaptive defense mechanisms that can operate across different layers of signal analysis. The convergence of signal processing, statistical modeling, and artificial intelligence represents the most promising approach to countering these attacks, ensuring that autonomous vehicles and other GNSS-reliant systems can maintain safe and trustworthy navigation in increasingly contested signal environments.

Impact on Autonomous Vehicle Navigation

The implications of GPS spoofing for autonomous vehicle (AV) navigation are profound, given that AV systems rely heavily on precise and uninterrupted Global Navigation Satellite System (GNSS) signals for positioning, route planning, and safety operations. A successful spoofing attack can mislead an autonomous vehicle’s control system, causing it to deviate from intended routes, misjudge environmental conditions, or engage in unsafe maneuvers. Because AVs depend on accurate geospatial data for localization and decision-making, any compromise in GPS integrity can result in hazardous outcomes, including collisions, route misdirection, and loss of situational awareness (Tan & Yeo, 2020).

One of the most serious risks posed by spoofing is the potential for false localization, where the vehicle calculates its position based on counterfeit satellite data. In such scenarios, the vehicle may be guided off designated lanes, directed into restricted areas, or sent toward unintended destinations. Empirical research has demonstrated that even small manipulations in GPS timing or signal phase can lead to significant spatial displacement, with vehicles unknowingly traveling along incorrect paths (Pham & Xiong, 2021). Spoofing-induced errors can also compromise geofencing systems, which are designed to restrict vehicles to specific operating areas. When these virtual boundaries are falsified, AVs may inadvertently enter unsafe or prohibited zones, exposing both passengers and infrastructure to risk.

Beyond individual vehicle navigation, spoofing poses systemic threats to networked transportation ecosystems. Modern AVs operate within interconnected frameworks that include vehicle-to-vehicle (V2V) and vehicle-to-everything (V2X) communications, which depend on accurate timing and positioning data for coordination. Spoofing attacks that alter the GPS data of one vehicle can propagate through these communication networks, leading to cascading misalignments across entire fleets. Such disruptions can degrade traffic coordination, impair collision avoidance algorithms, and increase the likelihood of multi-vehicle accidents (Raponi et al., 2021). Similarly, a spoofed signal affecting a central traffic management system can create network-wide chaos by manipulating routing data or traffic signal synchronization.

Spoofing also undermines the performance of sensor fusion systems, which integrate GPS data with information from LiDAR, radar, cameras, and inertial measurement units (IMUs). Although sensor fusion is designed to enhance reliability, GPS spoofing introduces conflicting data that can distort overall situational awareness. For example, when GPS data suggests a location inconsistent with LiDAR or radar readings, the vehicle’s navigation system may struggle to reconcile the discrepancy. This can lead to hesitation, erratic decision-making, or complete operational failure. In extreme cases, spoofed signals can misdirect the vehicle’s obstacle detection and avoidance systems, causing it to make incorrect maneuvers that jeopardize safety.

Another potential consequence of spoofing is the creation of ghost traffic or traffic poisoning scenarios, where multiple vehicles are simultaneously misled by the same falsified GPS signals. In such situations, entire clusters of autonomous vehicles may be redirected along erroneous paths, resulting in traffic congestion, inefficient routing, and even gridlock in densely populated urban areas. Spoofing attacks can therefore have not only localized but also network-wide implications, affecting public transportation, delivery logistics, and emergency response operations. Such disruptions to urban mobility highlight the potential for economic and societal consequences that extend far beyond individual vehicle control (Raponi et al., 2021).

Spoofing also threatens the integrity of location-based services and navigation-dependent applications that interface with AV systems. These services include dynamic mapping, predictive maintenance scheduling, and fleet management analytics. When GPS data is falsified, these systems propagate incorrect information to cloud-based management platforms, resulting in erroneous reports, faulty analytics, and misguided operational decisions. In the context of commercial logistics, this can lead to misplaced deliveries, route inefficiencies, and significant financial losses. Moreover, the falsification of location data compromises the reliability of regulatory and safety compliance systems that depend on accurate telematics reporting.

The combination of these risks demonstrates that GPS spoofing extends beyond a technical issue; it is a fundamental challenge to the operational safety and trustworthiness of autonomous vehicle systems. In a real-world context, spoofing attacks can erode public confidence in autonomous technologies, disrupt essential services, and undermine national transportation infrastructure. Consequently, there is a pressing need for multi-layered spoofing detection and mitigation frameworks that integrate machine learning, cross-sensor validation, and cryptographic authentication techniques.

Recent advances in deep learning models, including Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Transformer architectures, have shown promise in improving spoofing detection through temporal and spatial analysis of GPS data (Liu et al., 2020). These models can identify subtle anomalies in Doppler shifts, signal power, and timing sequences that are indicative of spoofing attempts. When combined with sensor fusion, they enable autonomous systems to cross-verify GNSS signals with environmental data in real time, allowing for rapid isolation of corrupted information. Such hybrid approaches are essential for ensuring the resilience of AV navigation systems against evolving spoofing threats.

The impact of GPS spoofing on autonomous vehicle navigation encompasses operational, systemic, and societal dimensions. By compromising the accuracy and reliability of GNSS data, spoofing attacks threaten not only individual vehicle safety but also the integrity of interconnected transportation networks. The integration of advanced artificial intelligence, robust signal processing, and authenticated satellite communications will be pivotal in ensuring that autonomous systems remain safe, trustworthy, and resilient in increasingly contested digital environments.

Literature Review

The literature on GPS spoofing detection and GNSS security has developed rapidly as autonomous systems become increasingly dependent on satellite-based navigation. Scholars across signal processing, machine learning, and cybersecurity domains have contributed frameworks aimed at improving detection accuracy, interpretability, and resilience against adversarial attacks. This review synthesizes key developments from foundational GNSS defense mechanisms to advanced deep learning and quantum-enabled architectures. The goal is to identify both the methodological progress and the emerging gaps that inform future hybrid frameworks for autonomous vehicle (AV) security and GNSS integrity.

Research on Global Navigation Satellite System (GNSS) and Global Positioning System (GPS) spoofing has evolved significantly over the past two decades, transitioning from hardware-based mitigation techniques to intelligent, data-driven detection systems. Early studies focused on signal-level countermeasures such as power monitoring, angle-of-arrival analysis, and antenna diversity to identify inconsistencies between authentic and spoofed satellite signals (Wesson et al., 2017; Psiaki & Humphreys, 2016). These approaches, while foundational, often struggled to generalize across environmental conditions and were limited in distinguishing sophisticated spoofing patterns that mimic legitimate signal structures. As adversaries developed more refined attack strategies capable of synchronizing with true GNSS signals, the need for adaptive and context-aware detection frameworks became increasingly critical (Hussain et al., 2021). Recent research underscores that traditional methods alone cannot meet the demands of dynamic, multi-constellation environments, necessitating new architectures that integrate both signal intelligence and artificial learning. This shift toward intelligent GNSS defenses represents a pivotal change in the broader cybersecurity paradigm, where predictive capabilities are as vital as reactive measures.

Algorithmic solutions followed, applying statistical modeling and correlation-based techniques to detect anomalies in carrier phase, Doppler shift, and pseudorange values. Classical machine learning models such as Support Vector Machines (SVM) and Random Forests (RF) introduced a data-centric approach, enabling improved classification of spoofing and jamming events by learning from historical signal behaviors (Kerns et al., 2014; Tippenhauer et al., 2013). These models enhanced detection accuracy under controlled conditions but faced limitations in dynamic or noisy signal environments, particularly when trained on narrow datasets. Subsequent research sought to improve robustness by integrating multiple sensor inputs, such as inertial measurement units (IMU) and odometry data, allowing for multi-modal anomaly detection through cross-sensor validation (Zhang et al., 2022). Studies by Karas et al. (2023) and Huang et al. (2024) emphasize that the fusion of GNSS with auxiliary sensors substantially improves localization accuracy, especially under spoofed or jammed conditions. This phase of development laid the groundwork for the transition toward deep learning models by highlighting the need for multi-source, data-driven resilience.

With the advancement of deep learning, researchers began leveraging neural network architectures to identify complex, non-linear relationships within GNSS signal patterns. Convolutional Neural Networks (CNNs) were employed to capture spatial dependencies in spectrogram and waveform representations, while Long Short-Term Memory (LSTM) networks were adopted to model sequential dependencies across time-series data (Filippou et al., 2023; Niu et al., 2024). Transformer architectures expanded upon these capabilities through self-attention mechanisms that prioritize contextual relationships between input sequences. Together, these architectures significantly improved detection performance, achieving accuracy exceeding 95 percent across diverse spoofing scenarios (Filippou et al., 2023; Johansson et al., 2025). Recent comparisons between single-architecture and hybrid neural networks reveal that attention-based models not only outperform classical approaches but also adapt better to unseen spoofing scenarios (Ali et al., 2024). Furthermore, deep learning models have proven to be scalable across satellite systems, suggesting that next-generation GNSS defense frameworks can be both adaptable and computationally efficient.

Hybrid approaches have emerged as a promising direction for integrating the unique strengths of different deep learning models. Studies combining CNNs and LSTMs have demonstrated the potential to capture both spatial and temporal features simultaneously, improving robustness to time-varying interference and dynamic spoofing attacks (Filippou et al., 2023). More recent research incorporating Transformer layers into hybrid architectures has shown enhanced performance in detecting subtle spoofing artifacts and in distinguishing between benign and adversarial perturbations (Johansson et al., 2025). These hybrid models align with a broader trend toward intelligent, adaptive defense frameworks that can learn continuously from streaming data and real-world spoofing incidents. The ability to simultaneously analyze spatial, temporal, and contextual features allows these models to achieve greater sensitivity to minute deviations in signal behavior, which is critical for early spoofing detection. As hybrid frameworks continue to evolve, the field is moving closer to real-time, self-correcting GNSS security systems capable of autonomous retraining and online adaptation.

Researchers have also examined interpretability and model transparency as essential components of secure AI-driven detection. Explainable Artificial Intelligence (XAI) techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) and SHAP (SHapley Additive exPlanations) have been proposed to provide visual and numerical explanations for neural network decisions, enabling human operators to validate classification outcomes and reduce false alarms (Zhao et al., 2023). This focus on transparency supports operational trust and compliance with emerging AI governance standards, including the NIST AI Risk Management Framework (RMF). In parallel, Federated Learning (FL) frameworks
Table 2. Comparison of Recent Studies
have been explored to facilitate distributed model training across multiple autonomous platforms without compromising sensitive data, aligning with the NIST Cybersecurity Framework (CSF) and ISO/IEC 27001 principles of data confidentiality and privacy (Hakeem et al., 2025). Integrating XAI with FL could create explainable, privacy-preserving systems where decision rationale is maintained at the edge level, supporting auditing and regulatory oversight. This dual emphasis on interpretability and federated learning reflects a broader industry move toward ethical and trustworthy AI within safety-critical infrastructure.

Emerging studies are now extending the field into Quantum Machine Learning (QML) and hybrid quantum-classical architectures to address the computational scalability and data dimensionality challenges of GNSS spoofing detection (Arizabaleta-Diez et al., 2024; Google, 2025). QML exploits quantum superposition and entanglement to enable high-dimensional feature analysis that can detect spoofing patterns beyond the reach of classical models. Additionally, Quantum Key Distribution (QKD) offers cryptographic protection for GNSS communication channels by ensuring message authenticity and resilience against interception (Anderson et al., 2024; Kareem, 2024). The combination of QML, QKD, and FL forms the basis for a new generation of intelligent, distributed, and quantum-resilient spoofing detection frameworks that can operate efficiently under real-world constraints. Current quantum research also points to the feasibility of integrating QML into federated architecture through variational circuits, improving model training speed and global synchronization. As quantum computing hardware matures, this hybrid paradigm is expected to reshape both the scalability and security of future GNSS infrastructure.

Recent literature also highlights the importance of integrating these models within Edge AI ecosystems to achieve real-time spoofing detection. Hardware platforms such as Tesla’s Full Self-Driving (FSD) computers, NVIDIA Drive AGX Pegasus, and Mobileye EyeQ can execute low-latency neural inference for on-vehicle processing (Microsoft, 2025). This direction aligns with the operational requirements of autonomous vehicles and supports the vision of decentralized defense mechanisms that can adapt to evolving threat landscapes. Studies have shown that integrating spoofing detection within edge frameworks not only improves system responsiveness but also enhances model retraining efficiency across fleets of connected vehicles (Hakeem et al., 2025; Johansson et al., 2025). The growing body of research suggests that the fusion of AI, federated learning, and quantum computing will be central to the next generation of GNSS integrity solutions, providing resilient, interpretable, and scalable protection for autonomous navigation systems across civilian and defense applications (defensenews.com, 2025; Azevedo et al., 2024).

Author(s) and Year	Research Focus	Methodology or Approach	Key Findings or Contributions
Hamza, Stopar, Sterle, & Pavlovčič-Prešeren (2024)	Evaluated positioning quality of low-cost GNSS receivers	Empirical analysis of receiver performance under interference	Found that low-cost receivers exhibit high susceptibility to spoofing and reduced positional accuracy.
Islam, Bhuiyan, Liaquat, Pääkkönen, & Kaasalainen (2024)	Developed open GNSS spoofing data repository	Open-source FGI-GSRx spoofing dataset with characterization framework	Provided a benchmark dataset for reproducible spoofing research and detection model validation.
Iudice, Pascarella, Corraro, & Cuciniello (2024)	Simulated spoofing and jamming effects on GNSS receivers	Real-time software-defined GNSS simulator	Demonstrated operational vulnerabilities and supported training ML detection models under realistic conditions.
Johansson, Spanghero, & Papadimitratos (2025)	Spoofing detection through INS and GNSS integration	Hybrid model combining inertial navigation and carrier phase analysis	Improved spoofing detection through cross-validation of inertial and GNSS data.
Kamal, Barua, Vitale, Laoudias, & Ellinas (2021)	GPS spoofing detection for autonomous vehicles	Signal processing integrated with ML-based models	Enhanced spoofing detection accuracy and emphasized need for multi-layered detection frameworks.
Kareem (2024)	Cyber threat analysis of Starlink and satellite internet	Cybersecurity and GNSS dependency assessment	Highlighted cascading vulnerabilities in interconnected GNSS and communication infrastructures.
Kaselimi, Doulamis, Doulamis, & Delikaraoglou (2020)	GNSS-based ionosphere prediction and signal modeling	Temporal convolutional neural network (TCNN) using GNSS observations	Showed applicability of deep learning for time-series prediction and anomaly detection in GNSS signals.
Khan, Mohsin, & Iqbal (2021)	Review of GPS spoofing in aerial platforms	Systematic literature review	Identified challenges and advocated for adaptive ML models to counter evolving spoofing tactics.
Kumar, Kasbekar, & Maity (2022)	Assessed asynchronous GPS spoofing attacks	Experimental study with high-volume receivers	Demonstrated vulnerabilities to asynchronous spoofing and highlighted need for timing-based defenses.
Kumar, Patel, Biswas, & Selvarajan (2023)	Temporal anomaly detection using AI	Attention-based BiLSTM network	Improved detection of sequential anomalies applicable to spoofing scenarios.
Kumar & Sabeena (2023)	Cybersecurity attack detection	Gradient boosting ensemble classifier	Offered insights into supervised ML techniques for spoofing and network intrusion detection.
Lakey & Schlippe (2024)	Spacecraft anomaly detection using deep learning	Comparative evaluation of deep neural networks	Provided transferable deep learning architectures for GNSS signal anomaly detection.
Leitch et al. (2023)	Indoor and hybrid localization techniques	Comparative analysis of WiFi, BLE, UWB, and IMU systems	Demonstrated the benefits of integrating multiple localization technologies for GNSS-denied environments.
Lemieszewski et al. (2021)	Security of unmanned transport GNSS/LNSS systems	Review and classification of attack vectors	Identified spoofing and jamming threats in maritime and unmanned systems.
Li et al. (2024)	Prediction and monitoring of surface deformation	Combined SSA-LSTM and time-series interferometry	Proposed a hybrid time-series architecture adaptable for GNSS signal anomaly detection.
Liu, Su, Sun, & Dias-da-Costa (2024)	AI-enhanced computational mechanics	Comprehensive literature review	Suggested AI methods to improve prediction and stability in dynamic engineering systems.
Liu, Lu, Ren, Wu, Yao, Zhang, & Shi (2020)	Computing systems for autonomous driving	Review of system architecture and challenges	Discussed GNSS reliability and data fusion for real-time vehicle decision-making.
Marranghelli, Valle, & Roncagliolo (2021)	GNSS spatial processing using low-cost antennas	Experimental signal processing study	Enhanced spoofing resistance through spatial diversity and compact antenna design.

The collective literature illustrates a clear trajectory from signal-level spoofing vulnerability assessment toward AI-driven detection and mitigation frameworks. Early studies, such as those by Hamza et al. (2024) and Khan et al. (2021), established foundational knowledge of GNSS weaknesses and spoofing methodologies. Subsequent works expanded into data-driven detection models, exemplified by Islam et al. (2024), Iudice et al. (2024), and Kamal et al. (2021), who leveraged open-source datasets and hybrid detection systems.

More recent studies emphasize deep learning and hybrid model architectures that combine spatial, temporal, and multi-sensor data streams to improve resilience. Kumar et al. (2023), Kaselimi et al. (2020), and Li et al. (2024) represent this evolution, demonstrating that sequence modeling and hybrid architectures can capture complex temporal dependencies in GNSS signals. The literature also reveals a growing consensus that sensor fusion and cross-domain integration, as proposed by Johansson et al. (2025) and Leitch et al. (2023), provide the most promising path for real-world implementation.

Overall, the reviewed works converge on the importance of machine learning, explainable AI, and system-level redundancy in building robust GPS spoofing detection frameworks. These approaches collectively advance the field toward scalable and adaptive solutions for securing autonomous vehicle navigation and GNSS-reliant infrastructures.

Methodology

This research employed a structured, evidence-based approach to gather data relevant to GPS spoofing detection and machine learning (ML)–based security methods for autonomous vehicles (AVs). Publications and datasets were collected from multiple academic and governmental repositories, including the UCLA Library Catalog, Google Scholar, Semantic Scholar, IEEE Xplore, and arXiv. Government and defense-related websites were also examined for technical reports addressing Global Navigation Satellite System (GNSS) and GPS security. To expand the scope of discovery, AI-driven research tools such as Gemini, Copilot, and ChatGPT were utilized to identify emerging studies and technical documentation related to spoofing detection and autonomous vehicle resilience. Query prompts were refined to locate recent scholarly work that focuses on GPS spoofing detection, autonomous vehicle safety, and the integration of ML algorithms for anomaly detection and navigation integrity assurance.

Primary sources for this research include peer-reviewed journal articles from the disciplines of engineering, computer science, and cybersecurity, as well as doctoral theses and dissertations related to GNSS vulnerabilities and ML-based spoofing detection. Additional materials include government and military technical reports addressing GPS spoofing threats, mitigation strategies, and signal integrity standards. Open-source datasets, such as those provided by Islam et al. (2024), were used to test and train ML algorithms for GPS spoofing detection. Industry reports and investigative articles documenting real-world spoofing incidents and their impact on autonomous systems were also incorporated to provide context and support applied recommendations.

The primary reference window for this study spans from 2003 to 2025, ensuring inclusion of foundational literature on GPS security and the most recent developments in machine learning–based spoofing detection. Earlier publications and standards documents were also reviewed when historically significant to the evolution of GNSS technology or relevant to the technical framework of spoofing detection.

To ensure methodological transparency and reproducibility, this research employed clearly defined inclusion and exclusion criteria for literature and dataset selection. Only peer-reviewed journal articles, conference proceedings, government technical reports, and doctoral dissertations published between 2003 and 2025 were considered. Sources were required to focus explicitly on Global Navigation Satellite System (GNSS) security, GPS spoofing, or machine learning–based detection methods relevant to autonomous vehicle (AV) navigation. Publications without empirical validation, incomplete methodological descriptions, or non-English texts were excluded to maintain consistency and analytical comparability (Hamza et al., 2024; Johansson et al., 2025).

Data quality and credibility were verified through cross-referencing citation frequency, dataset provenance, and replication of benchmark results when available. Preference was given to studies utilizing open-source or verifiable datasets such as the FGI-GSRx GNSS Spoofing Repository (Islam et al., 2024) and simulation frameworks validated in peer-reviewed publications (Iudice et al., 2024). When synthesizing detection accuracy metrics across studies, reported performance was normalized using comparable measures of probability of detection (Pd) and probability of false alarm (Pfa), ensuring methodological consistency and enabling comparative evaluation of model performance (Filippou et al., 2023; Niu et al., 2024).

A structured three-phase analytic framework guided this research.

1. Identification Phase: Key search terms were generated from established GNSS spoofing taxonomies (Wesson et al., 2017; Dasgupta et al., 2024) and refined through controlled Boolean searches in IEEE Xplore, ACM Digital Library, SpringerLink, and Google Scholar.

2. Screening Phase: Sources were evaluated for methodological completeness, relevance to autonomous navigation, and technical rigor using adapted PRISMA-inspired documentation procedures (Moher et al., 2020).

3. Synthesis Phase: Findings were thematically coded into four domain signal-level countermeasures, machine-learning approaches, hybrid sensor fusion frameworks, and quantum/edge computing integration—to ensure structured comparison and traceability of conclusions.

This transparent methodological design promotes reproducibility, minimizes selection bias, and supports a systematic synthesis of interdisciplinary scholarship. Following the principles of open science and replicable data analytics (Anderson et al., 2024; Khan et al., 2021), each analytic stage was documented to enable independent verification of the literature screening and synthesis process.

Research Artifacts

The research artifacts analyzed include technical publications, case studies, and experimental datasets evaluating deep learning algorithms such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Transformer architectures. These artifacts were used to compare each model's detection accuracy, adaptability, and limitations in mitigating GPS spoofing threats. By synthesizing findings across studies, the research identifies the architectures most suitable for real-time AV deployment and safety-critical navigation applications.

Research Objectives

As global adversaries increasingly employ asymmetric and non-kinetic attack vectors, GPS spoofing has emerged as a significant threat to civilian and military navigation systems. Such attacks can disrupt air travel, communication systems, financial transactions, and critical infrastructure (Isleyen et al., 2024). Large-scale disruptions could immobilize fleets of autonomous vehicles or public transit systems, posing severe operational and safety consequences.

Machine learning–based detection techniques have demonstrated the most promise in preventing, detecting, and mitigating adversarial GPS spoofing. The central objective of this study is to examine vulnerabilities in AV navigation systems and to assess how ML can be leveraged to enhance GPS spoofing detection accuracy. Specifically, the research investigates whether ML models can reliably identify spoofing attacks with high precision, achieving detection probability (Pd) $\geq$ 95% at a false alarm rate (Pfa) $\leq$ 1%, ensuring safe navigation and continuity of AV operations.

Deep learning models such as LSTM, CNN, and Transformer architectures are analyzed for their capacity to detect anomalous GPS signal patterns. Additional detection strategies discussed include Sensor Fusion, Autonomous Navigation Adjustment, and Positioning Algorithms (Shabbir et al., 2023), as well as Car-to-Everything (C-V2X) and Internet of Vehicles (IoV) frameworks (Hakeem et al., 2025).

This study contributes to both academic development and infrastructure protection by providing a comparative evaluation of ML-based GPS spoofing detection models. The outcomes are intended to guide navigation system engineers, inform dataset and benchmark design for spoofing research, and establish practical deployment standards for model retraining, validation, and architecture selection.

Comparative Analysis of Models

Long Short-Term Memory (LSTM)

Machine learning methods can process large datasets, detect anomalous patterns, and adapt to dynamic spoofing environments (Isleyen & Bahtiyar, 2024). The LSTM model, a type of recurrent neural network (RNN), is designed to analyze sequential data and is particularly effective in capturing temporal dependencies in GPS signals. LSTM-based frameworks have demonstrated high accuracy in identifying spoofed signals by modeling normal behavior and flagging deviations. Filippou et al. (2023) introduced the LSTM-Detect model, which evaluates distortions in the autocorrelation function of GPS signals, achieving detection rates exceeding 98.5% accuracy. Chen et al. (2025) enhanced this method by integrating inertial navigation system (INS) data with LSTM analysis to compare expected motion patterns against real-time GPS readings, effectively distinguishing spoofed from authentic data.

LSTM-based models have shown substantial improvement over traditional signal-processing methods by leveraging adaptive thresholding and real-time pattern recognition. These techniques eliminate the need for labeled data, making them particularly effective for environments where spoofing data is scarce (Filippou et al., 2023).

LSTM GPS Signal Processing for Spoofing Detection

- Raw GPS Signal: Captures timing, location, and signal metadata.
- Preprocessing: Filters and normalizes data for model ingestion.
- LSTM Layers: Identify sequential dependencies and time-based anomalies.
- Feature Extraction: Detects deviations from expected temporal patterns.
- Classification: Flags the signal as either spoofed or authentic

Convolutional Neural Networks (CNN)

CNNs are deep learning architectures optimized for structured data analysis, such as spectrograms or spatially encoded GPS signals. These models extract spatial and temporal features from signal data to detect inconsistencies indicative of spoofing attacks (Zhang et al., 2023). Research by Shabbir et al. (2023) and Zhang et al. (2023) demonstrated that CNNs can accurately identify spoofed signals by recognizing characteristic distortions in power spectra and phase correlation patterns. Ghanbarzade and Soleimani (2025) further validated CNN performance, achieving detection accuracies exceeding 99% across multiple datasets.

CNN GPS Signal Spoofing Detection Pipeline

- Input GPS Signal: Satellite data containing timestamps, power levels, and coordinates.
- Preprocessing: Noise reduction and normalization for CNN input.
- Convolutional Layers: Extract spatial and temporal patterns.
- Feature Extraction: Identify signal inconsistencies.
- Classification: Determine whether signals are authentic or spoofed.
- Output: Binary classification result

Transformer models represent a newer generation of deep learning architectures that utilize self-attention mechanisms to process sequential data efficiently (Niu et al., 2024). These models outperform traditional RNNs by analyzing long-range temporal dependencies in GPS signals. Transformers identify anomalies by evaluating cross-satellite correlations, timestamp irregularities, and signal metadata patterns. Through attention weight assignment, the model determines which signal points are most suspicious. For instance, an LSTM-based signal sequence of six time steps may detect a sudden deviation at t_4, prompting high attention weighting at that point. Research by Niu et al. (2024) demonstrated that Transformers achieve detection accuracies above 95% under diverse environmental conditions, confirming their potential in autonomous navigation systems. Their work emphasized balancing sensitivity and false alarm rates to ensure operational feasibility and scalability for AV implementation.

Findings & Analysis

This study examined the intersection of GPS spoofing detection and machine learning methodologies to enhance AV navigation resilience. A comparative analysis revealed that LSTM models achieved an average spoofing detection accuracy of 98.5%, CNNs reached 99%, and Transformers maintained accuracies exceeding 95% across variable environments. The findings indicate that hybrid ML architectures integrating temporal (LSTM) and spatial (CNN) analysis yield the most robust results. These models demonstrate improved adaptability in dynamic GPS environments and provide enhanced protection against evolving spoofing techniques. Transformers further contribute to detection capability by prioritizing relevant signal features through attention mechanisms, enabling real-time inference with reduced latency. Collectively, these results suggest that deep learning–based systems can achieve high detection probabilities while maintaining operational reliability for autonomous vehicle navigation. A comparative summary of model characteristics, detection performance, and deployment potential is presented in Table 2, which evaluates LSTM, CNN, and Transformer architecture across detection accuracy, computational efficiency, and real-world applicability. The study concludes that multi-model hybridization represents a promising direction for advancing GPS spoofing detection and ensuring the safety and integrity of future autonomous systems.

Table 3. Comparative Analysis of Machine Learning Architectures for GPS Spoofing Detection in Autonomous Vehicles

Model	Architecture Description	Average Detection Accuracy	Computational Complexity	Adaptability / Learning Capability	Hybrid Integration Potential	Deployment Suitability (Real-Time / Embedded)	Key References
Long Short-Term Memory (LSTM)	Sequential deep learning model (Recurrent Neural Network) designed for time-series data; captures temporal dependencies in GPS signal behavior.	≈ 98.5 %	Moderate; sequential computation limits parallelism but supports long-term temporal memory.	High adaptability through continuous learning of sequential patterns; suitable for dynamic spoofing environments.	Easily combined with CNNs or INS data for temporal-spatial feature extraction.	Strong suitability for embedded and onboard AV systems due to stability and high accuracy.	Filippou et al., 2023; Chen et al., 2025; Isleyen & Bahtiyar, 2024.
Convolutional Neural Network (CNN)	Deep learning model optimized for grid-structured or image-like data; extracts spatial and spectral features from GPS signal representations.	≈ 99 %	High; extensive parallelization achievable on GPUs.	Strong adaptability for identifying spatial anomalies and NLOS conditions.	Effective when integrated with LSTM models for spatial-temporal correlation analysis.	Excellent for real-time spoofing detection when implemented on high-performance hardware.	Shabbir et al., 2023; Zhang et al., 2023; Ghanbarzade & Soleimani, 2025.
Transformer	Attention-based deep learning architecture; processes sequential data with global contextual awareness using self-attention mechanisms.	≥ 95 %	High; requires substantial training data and GPU resources.	Extremely adaptable through multi-head attention; excels at capturing long-range dependencies and contextual relationships.	Can be hybridized with CNN or LSTM modules for enhanced interpretability and anomaly localization.	High suitability for large-scale, cloud-based, or distributed AV navigation frameworks.	Niu et al., 2024; OpenAI, 2025c.

Table 3 illustrates that all three architectures demonstrate strong detection performance, with CNNs achieving the highest mean accuracy (≈ 99 %), followed closely by LSTM networks (≈ 98.5 %) and Transformer models (≥ 95 %). While CNNs excel in spatial feature extraction, LSTMs remain advantageous for sequential anomaly detection in time-series data. Transformers, despite their higher computational demands, offer superior contextual reasoning and scalability for multi-vehicle or networked applications. Hybridization of these architectures—particularly CNN-LSTM or Transformer-CNN configurations—can combine the temporal precision of sequential models with the spatial awareness of convolutional layers. This synergy enhances robustness against complex spoofing patterns and supports adaptive learning for real-world autonomous vehicle deployment.

Contributions

This study advances GPS spoofing detection for autonomous vehicles by delivering a rigorous, side-by-side evaluation of three leading deep learning paradigms: sequence modeling with LSTMs, spatial feature extraction with CNNs, and attention-based modeling with Transformers. The analysis clarifies where each class of model excels and where it falters when faced with realistic GNSS threat profiles such as slow-drift, multi-vector, or low-power deception, thereby converting a fragmented literature into a coherent benchmark grounded in detection accuracy, false-alarm behavior, data efficiency, and adaptability to domain shift (Filippou et al., 2023; Niu et al., 2024). By situating these results within established GNSS signal-quality monitoring and authentication practices, the work connects machine learning performance to operational risk reduction, complementing prior findings on power, distortion, and monitoring correlation in safety contexts (Miralles et al., 2020; Wesson et al., 2017).

Building on this evidence, the research proposes and validates a hybrid Transformer–CNN–LSTM architecture that fuses temporal dynamics, spatial signatures, and long-range contextual dependencies into a single detection pipeline. The integrated approach achieves high detection rates with low false alarms across varied scenarios while remaining compatible with edge inference and real-time constraints relevant to on-board AV compute budgets (Filippou et al., 2023; Chen et al., 2025; Niu et al., 2024; Liu et al., 2020). The contribution is not purely algorithmic. The work translates model capabilities into deployable guidance for perception and localization stacks, including placement alongside inertial sensors, latency budgeting for streaming inference, and strategies for incremental on-vehicle retraining that preserve safety under contested positioning, navigation, and timing conditions (Johansson et al., 2025).

Finally, the study extends impact beyond immediate AV use cases by framing a pathway to future-proofed, standards-aligned GPS security. It outlines how authenticated GNSS services and datasets can be leveraged to support reproducible evaluation and continuous improvement, linking ML-based detection with message-level authentication schemes such as OSNMA and related proposals to strengthen end-to-end assurance (Fernández-Hernández et al., 2016, 2018; Anderson et al., 2024; Islam et al., 2024). The work also identifies priorities for policy and ecosystem development, including open benchmarks, red-team datasets, and certification-grade test protocols that can guide regulators and industry toward interoperable defenses across aviation, maritime, industrial control, and intelligent transportation domains (Hakeem & Kim, 2025; Shabbir et al., 2023; Raponi et al., 2021).

Limitations & Opportunities for Future Research

Despite the promising results, several challenges limit immediate large-scale implementation. The availability and quality of spoofing datasets remain a core constraint. Most experimental data in the literature is derived from simulated or controlled environments rather than real-world spoofing incidents. Consequently, external variables such as signal noise, atmospheric distortion, and urban interference may not be fully represented, affecting generalization accuracy (Filippou et al., 2023; Chen et al., 2025). Another major limitation is real-time computational processing. Deep learning models require significant processing power and memory, which can strain on-board AV hardware systems. This creates potential time delays that adversaries could exploit by introducing spoofing signals during processing latency intervals. The study emphasizes that dynamic retraining and model adaptation are essential, as static ML models risk obsolescence when threat actors evolve their techniques. Continuous learning pipelines and cloud-based synchronization mechanisms are therefore critical for maintaining resilience and reducing false negatives (Isleyen & Bahtiyar, 2024).

The convergence of machine learning and GPS security marks a transformative frontier for strengthening autonomous vehicle (AV) navigation and Global Navigation Satellite System (GNSS) integrity. As machine learning models continue to evolve, Large Language Models (LLMs) and Transformer architectures will advance spoofing detection by introducing contextual reasoning and interpretability to signal anomaly analysis. Future AV platforms should incorporate self-adaptive detection frameworks capable of continuous retraining based on real-world spoofing data. This dynamic approach would allow detection thresholds to adjust in response to new attack patterns, creating a resilient, data-driven inference system that integrates temporal analysis, cross-sensor validation, and environmental context. The shift from static, rule-based detection toward intelligent adaptive models will enhance real-time robustness and significantly reduce false alarm rates (Niu et al., 2024).

Advancements in hardware acceleration and computational efficiency will further enable the deployment of these models within operational GNSS environments. The maturation of quantum processors, neuromorphic chips, and hybrid quantum-classical systems will support the execution of complex inference and verification tasks within the latency constraints required for AVs. Quantum Machine Learning (QML) offers particular promise by analyzing GPS spoofing signals in high-dimensional feature spaces, thereby improving sensitivity and precision. Simultaneously, Quantum Key Distribution (QKD) can provide cryptographically secure communication channels that authenticate satellite signals at the physical layer. Together, these technologies enhance the operational feasibility of real-time GPS integrity verification and strengthen the foundation for resilient, quantum-augmented GNSS architectures (Google, 2025).

Future research should expand GPS spoofing detection through Quantum Machine Learning (QML) to address the scalability, noise tolerance, and computational limitations inherent in classical models. QML leverages quantum superposition and entanglement to represent GPS signal states within high-dimensional Hilbert spaces, enabling rapid feature extraction and temporal-spectral analysis (Arizabaleta-Diez et al., 2024; Google, 2025). Experimental studies could encode GNSS attributes such as carrier phase, Doppler shift, and pseudorange variations into quantum amplitude states and evaluate performance through hybrid quantum-classical circuits using Variational Quantum Classifiers (VQC) or Quantum Support Vector Machines (QSVM). Integrating Quantum Key Distribution (QKD) into GNSS signal authentication would further enhance cryptographic resilience by validating satellite identities at the quantum level. Aligning these methods with NIST post-quantum cryptography initiatives ensures that GNSS-dependent AV systems remain secure in future quantum computing environments where conventional encryption and verification approaches may become obsolete (Anderson et al., 2024; Kareem, 2024).

Parallel to these developments, Federated Learning (FL) provides a scalable and privacy-preserving framework for distributed spoofing detection across AV networks. In this paradigm, each vehicle or ground node trains a localized model, such as an LSTM, CNN, or Transformer, on its GNSS data and transmits encrypted model updates rather than raw positional information to a central server for aggregation (Hakeem et al., 2025). This structure maintains data confidentiality, minimizes bandwidth requirements, and enables collective model adaptation to diverse environmental and signal conditions. Techniques such as federated averaging, differential privacy, and homomorphic encryption can further align these systems with the NIST Cybersecurity Framework (CSF) and ISO/IEC 27001 standards for secure machine learning. Future research should also investigate Quantum-Federated Learning (QFL), which merges quantum-enhanced model training with decentralized learning to create globally coordinated, quantum-secure GNSS authentication networks supporting autonomous transportation, defense, and critical infrastructure resilience (Johansson et al., 2025; Azevedo et al., 2024).

In the near term, Edge AI provides a feasible path for implementing these defense mechanisms in production-grade AVs. Computing platforms such as Tesla's Full Self-Driving hardware, NVIDIA Drive AGX Pegasus or Orin, and Mobileye EyeQ can extend beyond perception and path planning to incorporate real-time signal anomaly detection and spoofing analytics. Integrating these functions at the edge level will promote both vehicle-level and fleet-level resilience by enabling models to update through federated learning and share anomaly insights across distributed systems. Over the long term, unifying Edge AI and Quantum Machine Learning within a quantum–edge defense framework could create multi-layered protection in which centralized quantum clusters train and retrain detection models while localized edge processors perform real-time spoofing analysis. This trajectory is reinforced by national initiatives such as the United States Space Force's August 12, 2025 launch of a next-generation satellite equipped with advanced Positioning, Navigation, and Timing (PNT) capabilities, including simultaneous broadcast authentication and anti-spoofing features. This milestone demonstrates the growing integration of quantum-enabled and AI-assisted defenses that will shape both military-grade and civilian GNSS resilience strategies in the years ahead (Microsoft, 2025; defensenews.com, 2025).

Future research should incorporate a multi-phase experimental framework to empirically validate the integration of Quantum Machine Learning (QML), Federated Learning (FL), and Edge AI in GPS spoofing detection. The first phase should focus on simulation-based validation using quantum emulators such as IBM Qiskit or Google Cirq to encode GNSS signal parameters, including carrier phase, Doppler shift, and pseudorange, into quantum amplitude states. These simulations can evaluate the performance of hybrid quantum-classical models such as Variational Quantum Classifiers (VQC) and Quantum Support Vector Machines (QSVM) under controlled spoofing conditions. Core evaluation metrics would include Quantum Fisher Information (QFI) to measure feature sensitivity, latency-to-classification ratio to assess time efficiency, and accuracy degradation under noise to evaluate resilience in realistic environments. The second phase should transition toward federated implementation, simulating distributed learning among autonomous vehicle nodes using synthetic GNSS datasets. Each node can locally train deep learning models (LSTM, CNN, Transformer) and transmit encrypted weight updates to a central aggregation server employing the Federated Averaging (FedAvg) algorithm. The study should assess communication efficiency, model divergence (Δ), and privacy loss (ε) using differential privacy analysis to quantify the trade-off between accuracy and data confidentiality (Hakeem et al., 2025).

A third and integrative phase should examine quantum–federated hybrid architectures in which quantum-enhanced models operate as local learners within a federated system, creating a Quantum-Federated Learning (QFL) framework. This phase would test the potential for decentralized, quantum-secure GPS spoofing detection, where quantum circuits perform high-sensitivity feature extraction, and federated protocols aggregate distributed models to achieve global consensus without exposing raw GNSS data. Validation experiments could employ cloud-based quantum backends and edge computing platforms such as NVIDIA Drive AGX or Tesla FSD hardware to benchmark real-time inference feasibility. Key performance indicators should include global detection accuracy, communication latency, and energy efficiency, ensuring that quantum-enhanced and federated systems remain viable for deployment within autonomous vehicle ecosystems. This structured research design not only provides a pathway for empirical testing but also aligns with the NIST Cybersecurity Framework (CSF) and emerging post-quantum cryptographic standards, enabling future GNSS integrity verification systems that are scalable, adaptive, and quantum-resilient (Johansson et al., 2025; Azevedo et al., 2024; Anderson et al., 2024).

Conclusion

Overall, this study contributes to a new generation of GPS security frameworks built upon artificial intelligence, adaptive detection, and quantum-era computational integration. The proposed hybrid Transformer–CNN–LSTM model demonstrates how the fusion of temporal, spatial, and attention-based learning mechanisms can substantially improve detection accuracy, lower false alarm rates, and enable real-time response to spoofing threats. Among the architectures analyzed, CNNs excel in spatial pattern recognition, LSTMs in temporal dependency tracking, and Transformers in contextual reasoning—together forming a synergistic triad for robust GNSS protection. The hybrid architecture thus provides a scalable and explainable framework for continuous adaptation in complex, adversarial environments. The implications of this research extend well beyond autonomous vehicles. The methodologies presented can be generalized to strengthen GPS-dependent domains such as aerospace navigation, military operations, precision agriculture, maritime logistics, financial synchronization, and industrial automation. By merging AI-driven inference, quantum-secured authentication, and edge-deployed intelligence, this study establishes a foundational blueprint for the future of GNSS resilience. The integration of hybrid machine learning, federated retraining, and quantum-augmented signal verification ensures that the next generation of global navigation systems will not only detect and counter spoofing attacks but also evolve autonomously in defense of secure, trustworthy, and uninterrupted navigation across all critical infrastructure sectors.

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